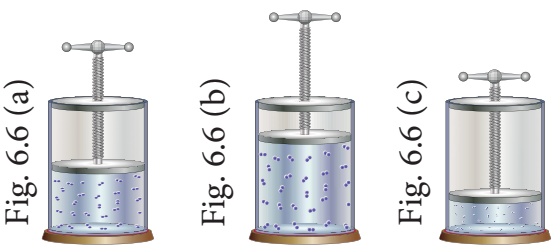
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If we extrapolate the straight line in the figure 6.5 beyond the experimental measurements, the straight line intersects the temperature axis (x-axis) at -273° C. This shows that the volume of the gas becomes zero at -273° C, more precisely this temperature is -273.15° C. Beyond this temperature the gas would have a negative volume which is physically impossible. For this reason, this temperature was defined as absolute zero by Kelvin and he proposed a new temperature scale with absolute zero as starting point which is now called Kelvin scale. The only difference between the Kelvin scale of temperature and Celsius scale of temperature is that the zero position is shifted. The boiling and freezing point of water in both scales are given below.

| | Kelvin Scale | Celsius scale |
|---|---|---|
| Absolute Zero | 0 K | -273.15 °C |
| Freezing point of water | 273.15 K | 0 °C |
| Boiling point of water | 373.15 K | 100 °C |

**Example:**

In figure 6.6 let us find the missing parameters (volume in 6.6 (b) and temperature in 6.6(c))




P = 1 atm. $V_1 = 0.3\ dm^3$ $T_1 = 200\ K$

P = 1 atm. $V_2 = ?\ dm^3$ $T_2 = 300\ K$

P = 1 atm. $V_3 = 0.15\ dm^3$ $T_3 = ?\ K$

**Fig. 6.6. Effect of temperature on volume of the gas to verify Charles law**

Solution:

According to Charles law,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{V_3}{T_3}$$

$$\frac{0.3\ dm^3}{200\ K} = \frac{V_2}{300\ K} = \frac{0.15\ dm^3}{T_3}$$

$$\frac{V_2}{300\ K} = \frac{0.3\ dm^3}{200\ K}$$

$$V_2 = \frac{1.5}{\cancel{300\ K} \times 0.3\ dm^3}{\cancel{200\ K}}$$

$V_2 = 0.45\ dm^3$ and

$$\frac{0.15 dm^3}{T_3} = \frac{0.3\ dm^3}{200\ K}$$

$$T_3 = \frac{0.5}{\cancel{0.15 dm^3} \times 200\ K}{\cancel{0.3\ dm^3}}$$

$T_3 = 100\ K$

**Evaluate Yourself**

**3.** A sample of gas has a volume of 3.8 $dm^3$ at an unknown temperature. When the sample is submerged in ice water at 0 °C, its volume gets reduced to 2.27 $dm^3$. What is its initial temperature?

### 6.2.3 Gay-Lussac's Law (Pressure-temperature relationship)

Joseph Gay-Lussac stated that, at constant volume the pressure of a fixed mass of a gas is directly proportional to temperature.

$P \alpha T$